# Kenneth W. Bingham

**Senior Full-Stack Developer | AI-Directed Engineer**

St. George, UT (Mountain Time) | Remote, or hybrid in St. George | Contact: butterflyfx.us/contact.html

LinkedIn: linkedin.com/in/kenneth-bingham-11378b10 | GitHub: github.com/kenbin64

Portfolio: butterflyfx.us | Games: playbutterflyfx.com | Gallery: kensartgallery.com | Videos: youtube.com/@kbingh | Substack: substack.com/@kbingh

## SUMMARY

Senior full-stack developer and AI-directed engineer with 15+ years building enterprise, government, financial and logistics systems in Java, Spring Boot, C#, .NET Core, Angular, SQL and Azure DevOps. Builds adversarial AI benchmarks and evaluation harnesses, and directs AI as an engineering lead with tests that try to break each claim.

## SKILLS

**Core:** Java, Spring Boot, Hibernate and JPA, C#, .NET 6 and .NET Core, Razor Pages, Angular, TypeScript, JavaScript, SQL Server, PostgreSQL, REST APIs and microservices, Python, Node.js

**AI engineering:** Claude Code, agentic workflows, MCP, RAG with pgvector, prompt engineering, LLM evaluation (pass@k), adversarial benchmarks, hallucination mitigation, Ollama

**DevOps and testing:** Azure DevOps, TFS, Git, CI/CD, Cloudflare Workers, Ubuntu and nginx administration, JUnit, Postman, Swagger, Puppeteer, debuggers

**Design patterns:** MVC, Singleton, Builder, Factory, Observer, publish-subscribe, dependency injection

**Previous experience:** AngularJS, jQuery, GWT, JSP, Oracle, MySQL, MongoDB, MarkLogic, Selenium, TestNG, AWS, Docker, Tomcat, TIBCO, Splunk, Adobe AEM

**Currently learning:** React, Next.js, Kotlin

## EXPERIENCE

**AI Engineer, Evaluation and Reliability (Contract)**

Handshake AI, Remote | Jul 2026 to Present

- Author adversarial benchmark tasks for Terminal-Bench 2 that defeat frontier coding agents, verified at pass@5 0/5: the reference agent at maximum reasoning effort fails every attempt.
- Built an offline difficulty-certification harness (candidate generator, CP-SAT constraint-solver filter and a blind cold-agent pass@2 run) that certifies task difficulty in minutes instead of a 68-minute CI cycle.
- Wrote a deterministic pre-submit linter encoding 104 authoring rules that catches solution leaks, ambiguous framing and self-verifiable traps before a task reaches review.

**Software Developer**

DTSI (defense contractor), Hill Air Force Base, UT | Apr 2024 to May 2026

- Principal developer on Air Force logistics and operational systems in Java, Spring Boot, Hibernate, Angular and Bootstrap, and in C#, .NET Core Razor Pages and SQL Server, under strict DoD change control.
- Configured CI/CD pipelines and secure deployment workflows with TFS version control, with zero rollbacks.

**Software Developer (Contract via Brooksource)**

Humana, Remote | Feb 2022 to Jan 2024

- Built .NET 6 services in C# and Visual Studio, and HL7 and FHIR interoperability pipelines for high-volume insurance claims under HIPAA.
- Modernized legacy applications from .NET Framework to .NET Core (.NET 6) for performance and cross-platform support.

**Software Developer**

Software Implementers (Air Force contractor), Hill Air Force Base, UT | Apr 2021 to Dec 2021

- Developed Java, Spring, Spring Boot and Hibernate services and responsive Angular and Bootstrap front ends for aircraft-parts testing, maintenance and asset tracking.
- Turned the testers' manual spreadsheets into a central web-based CMS and workflow platform for parts tracking, test protocols and maintenance.

**Software Developer (Contract via ConsultNet)**

Client: The Church of Jesus Christ of Latter-day Saints, Salt Lake City, UT | Sep 2020 to Oct 2020

- Contributed to an internal application for tracking and versioning Church leaders' talks, in Java, Spring Boot, Hibernate, JSP, Thymeleaf and Bootstrap.

**Software Developer (Contract via Collabera)**

Gwinnett County Public Schools, Lawrenceville, GA | Feb 2020 to Aug 2020

- Built automated test frameworks in Java and Selenium WebDriver for the district's cloud platforms, with Azure DevOps pipelines.

**Contract and agency roles**

Norfolk Southern and IBM (via Collabera), Finicity (Mastercard, via STG); InComm and WEX | 2017 to 2019

- Spring Boot and TIBCO messaging for locomotive fuel efficiency; Selenium WebDriver automation with grid parallel testing; payment and prepaid card systems; financial data aggregation microservices; a fleet gas-card portal.

**Java Developer**

CUProdigy, Layton, UT | Apr 2015 to Feb 2017

- Developed the credit union core system for small and mid-size credit unions: teller applications, management and operations, record keeping, account management, loan processing, interest calculation, and payment and collection management.

**Software Consultant**

Software Technology Group, Salt Lake City, UT | Sep 2011 to Jun 2014

- Missionary Portal for The Church of Jesus Christ of Latter-day Saints, used by thousands of missionaries worldwide: helped plan the architecture and built most of the UI (Bootstrap, Backbone, Handlebars) with Java Spring services and MarkLogic.
- A translation application for the same client (Java Swing, Flex), and CourtView, a case and docket application used by courts nationwide (Java, Spring, DataTables).

**Contract roles**

Axis41 via ConsultNet (Adobe AEM for O.C. Tanner and Stanford Health Care, Nov 2014 to Apr 2015); USANA Health Sciences via TEKsystems (AngularJS, Jun to Oct 2014) | Jun 2014 to Apr 2015

**Earlier experience**

Boresight (internship, Jun 2010 to Sep 2011: Java Swing avionics); Convergys (Juniper router technical support, 2008 to 2009); Delta Air Lines (reservations, 18 years) | Before 2011

## SELECTED PROJECTS (HUMAN AND AI COLLABORATION)

In each project I set the vision, design, standards and priorities and verified every result; an AI coding assistant wrote and tested the code under my direction.

- **Chomp 3D: Wally's Cairns Adventure, playbutterflyfx.com/chomp.** Seven-level 3D browser game for desktop, phone and VR (WebXR). 1,022 unit checks and 107 browser tests gate 14 versioned releases with one-command Cloudflare rollback; Workers leaderboard and feedback form.
- **citeline, github.com/kenbin64/citeline.** Retrieval-augmented generation over federal regulations with PostgreSQL, pgvector embeddings and FastAPI: answers with a citation or abstains. 31 unit tests and a 28-case evaluation set; a live version runs on Cloudflare Workers.

## EDUCATION

B.S., Computer Science and Software Engineering, Weber State University, Ogden, UT, 2011

## CERTIFICATIONS

CompTIA A+

## SECURITY CLEARANCE

U.S. Air Force Secret clearance (previously held), with years of delivery inside DoD environments at Hill Air Force Base.

## INTERESTS

3D printing, game development, AI art, drumming, painting, singing.